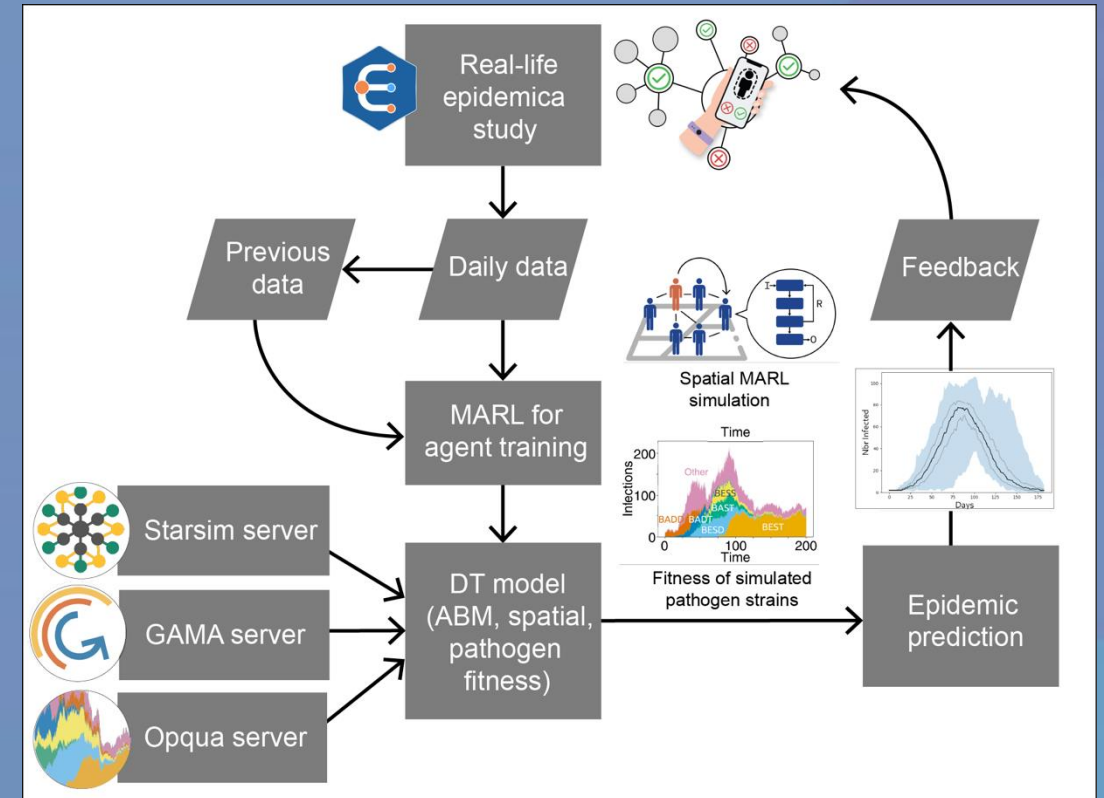


Epidemica:

An Open, Agentic-Ready Research Platform for Participatory Epidemiology Studies

Andrés Colubri, MFA PhD
Assistant Professor, UMass Chan Medical School



What's the problem?

Epidemiology research needs new toolkits for data collection and experimentation

- Many epidemiological models depend on contact patterns, but most use synthetic data — not measured behavior
- No platform lets researchers define a study, deploy it, and collect standardized behavioral + contact data — without writing an app from scratch
- Existing tools are siloed: you choose between contact tracing, behavioral surveys, or gamified engagement — never all three

Epidemica solves this by providing one open-source platform where proximity sensing, transmission simulation, surveys, and gamification are composable modules that any research group can assemble into a study

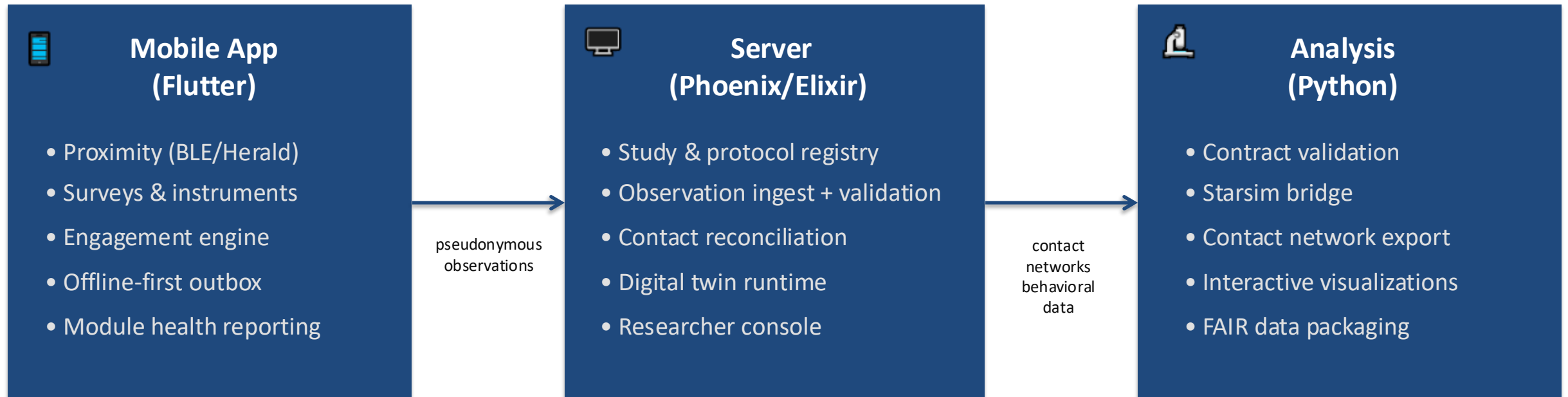
What is Epidemica?

An open-source framework for building participatory epidemiology studies

- Modular architecture: proximity sensing, transmission modelling, surveys, gamification — each is an independent, composable module
- Three tiers of effort:
 - Tier 1 — Write a JSON protocol bundle, no code, no app store review
 - Tier 2 — Custom app with bespoke screens and user interfaces (like Epigames)
 - Tier 3 — Build a new measurement module
- Self-hosted Phoenix/Elixir server + PostgreSQL — no cloud vendor lock-in
- GDPR-ready: all observation data is pseudonymous; PII is physically separable
- Universal data format: every module emits in the same "Observation Envelope" into one append-only store

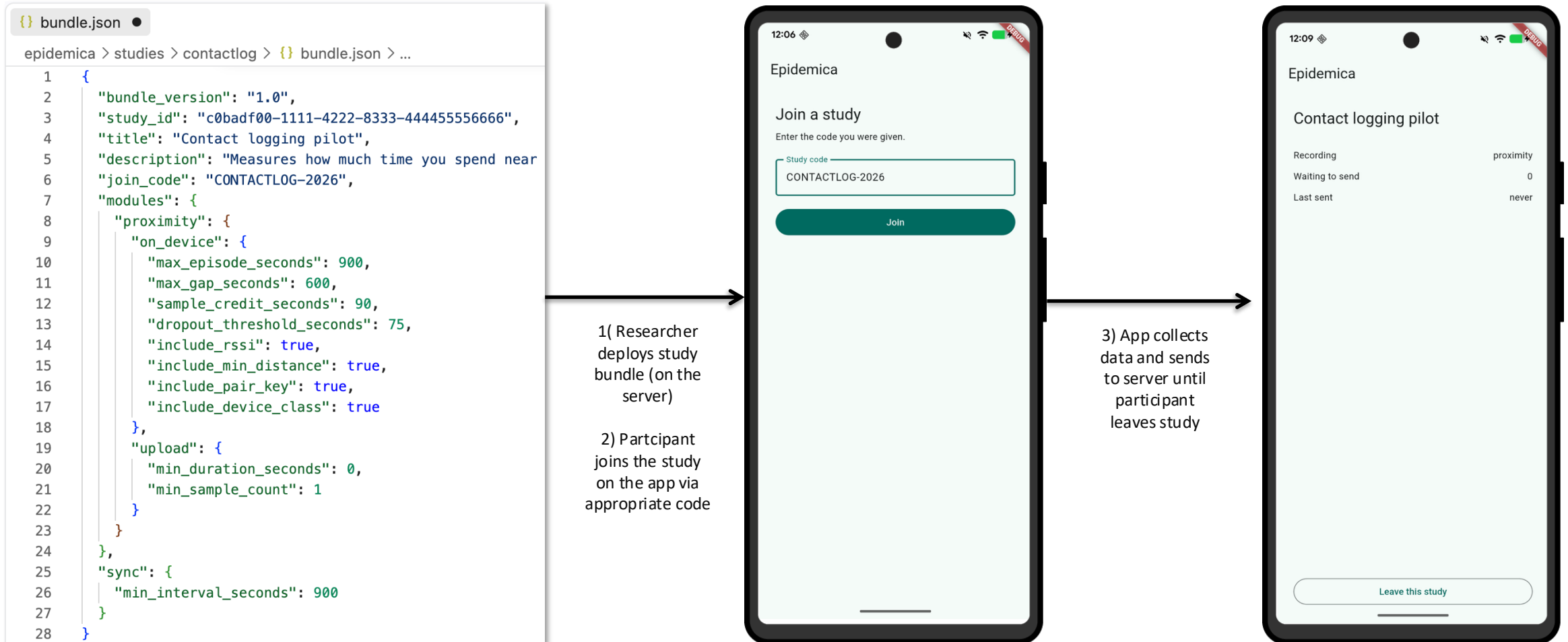
How it works

Three layers, one data flow



How it works

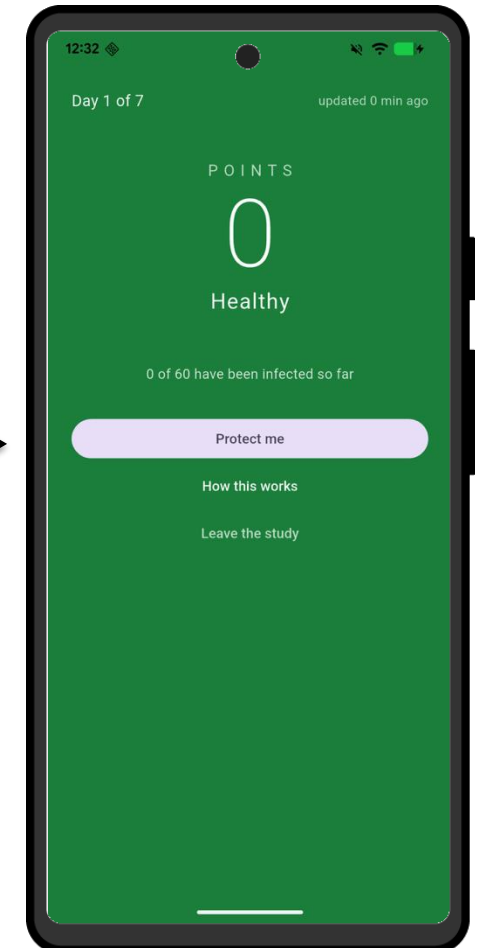
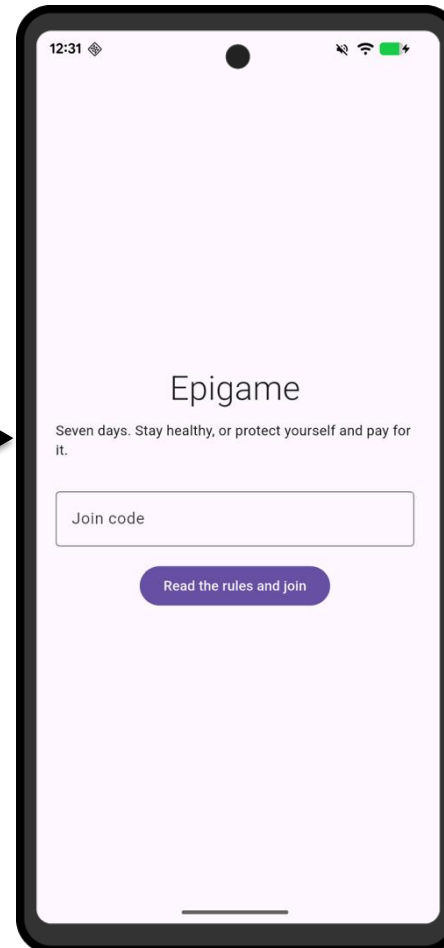
Configure the modules to be used in a study app via a bundle file (Tier 1)



How it works

Write app with custom user interface, still use bundle to load Epidemica modules (Tier 2)

```
bundle.json X
epidemica > studies > epigame-debug > {} bundle.json > {} rules > {} pars > ## healthy_points
1
2 {
3   "bundle_version": "1.0",
4   "study_id": "d3b06a7e-0007-4a11-8022-c3d4e5f6d3b0",
5   "title": "Epigame: debug",
6   "description": "A compressed run of the seven-day game, for checking that the pipeline works. Each round lasts
7   "join_code": "EPIGAME-DEBUG",
8   "modules": {
9     "proximity": {
10       "on_device": {
11         "max_episode_seconds": 900,
12         "max_gap_seconds": 600,
13         "sample_credit_seconds": 90,
14         "dropout_threshold_seconds": 75,
15         "include_rssi": true,
16         "include_min_distance": true,
17         "include_pair_key": true,
18         "include_device_class": true
19       },
20       "upload": {
21         "min_duration_seconds": 60,
22         "min_sample_count": 3
23       }
24     },
25     "survey": {
26       "instruments": [
27         {
28           "instrument_id": "checkin",
29           "version": "1.0.0",
30           "sha256": "sha256:d6e08a3032e14a45b7274ef30a8dc529227fc548bbb06462998bdcf77b429946",
31           "offset_seconds": 60,
32           "window_seconds": 1800
33         }
34       ]
35     },
36     "sync": {
37       "min_interval_seconds": 60
38     },
39     "health": {
40       "enabled": true,
41       "interval_seconds": 60
42     },
43     "schedule": {
44       "starts_at": "2026-09-03T06:00:00Z",
45       "days": 7
46     },
47     "twin": {
48       "engine": "starsim",
49       "state_url": "https://schemas.epidemica.info/state/epigame/1.0.0.json",
50       "tick_interval_seconds": 300,
51       "population": 60,
52       "seed": {
53         "infections": 3,
54         "among": "virtual"
55       },
56       "para": {
57         "diseases": {
58           "type": "sir",
59           "beta": 0.35,
60           "dur_inf_days": 5,
61           "dur_inf_std_days": 1.5
62         },
63         "virtual": {
64           "contacts_per_day": 6,
65           "band_seconds": {
66             "immediate": 300,
67             "close": 900,
68             "medium": 600,
69             "far": 0
70           }
71         },
72         "protection": {
73           "efficacy": 1.0,
74           "blocks_transmission": true
75         }
76       }
77     },
78   }
79 }
```



What we have built so far

Two milestones complete, from contract to simulation

Capability	Status	What it means
Observation Envelope + contracts	✓ Done	Universal data format — one ingest path for all module types
Proximity module (Herald/BLE)	✓ Done	Extracted into a proper Flutter plugin; distance estimation, episode aggregation, tested in CI
Phoenix/Elixir server	✓ Done	Enrolment, ingest, validation, contact projection, researcher console — single deployable
No-code study bundles (Tier 1)	✓ Done	JSON protocol → register → deploy. No app code, no store review per study
Digital twin (Starsim)	✓ Done	Server-side ABM with virtual participants completing the population; reproducible ticks
Epigames v2 app	✓ Done	Complete rewrite using the twin model; solves Bluetooth-off handling, protection scoring
Contact reconciliation	✓ Done	Two-sided contacts merged via interval union; deterministic, rebuildable
Data export + visualization	✓ Done	Contract-validated exports, Starsim network loading, interactive visualizations

The Digital Twin model

Server-side Starsim simulation driven by real contact data

- Real participants carry phones → BLE detects who is near whom → episodes are uploaded
- The server reconciles both sides of every contact (union of intervals, distance from better-observed side)
- Virtual participants fill the population to the declared size — 20 real players can drive a 60-person epidemic
- Starsim runs one tick per day: real contacts + virtual mixing → infections, recoveries, deaths
- Key breakthroughs in the new implementation:
 - Bluetooth off gracefully merged into game mechanics (participants were “kicked out” before)
 - Every infection is attributable to a specific contact or virtual event
 - Every tick is stored, reproducible, and re-runnable from its inputs
 - All scoring constants are bundle parameters — different game economics need no rebuild

The Digital Twin model

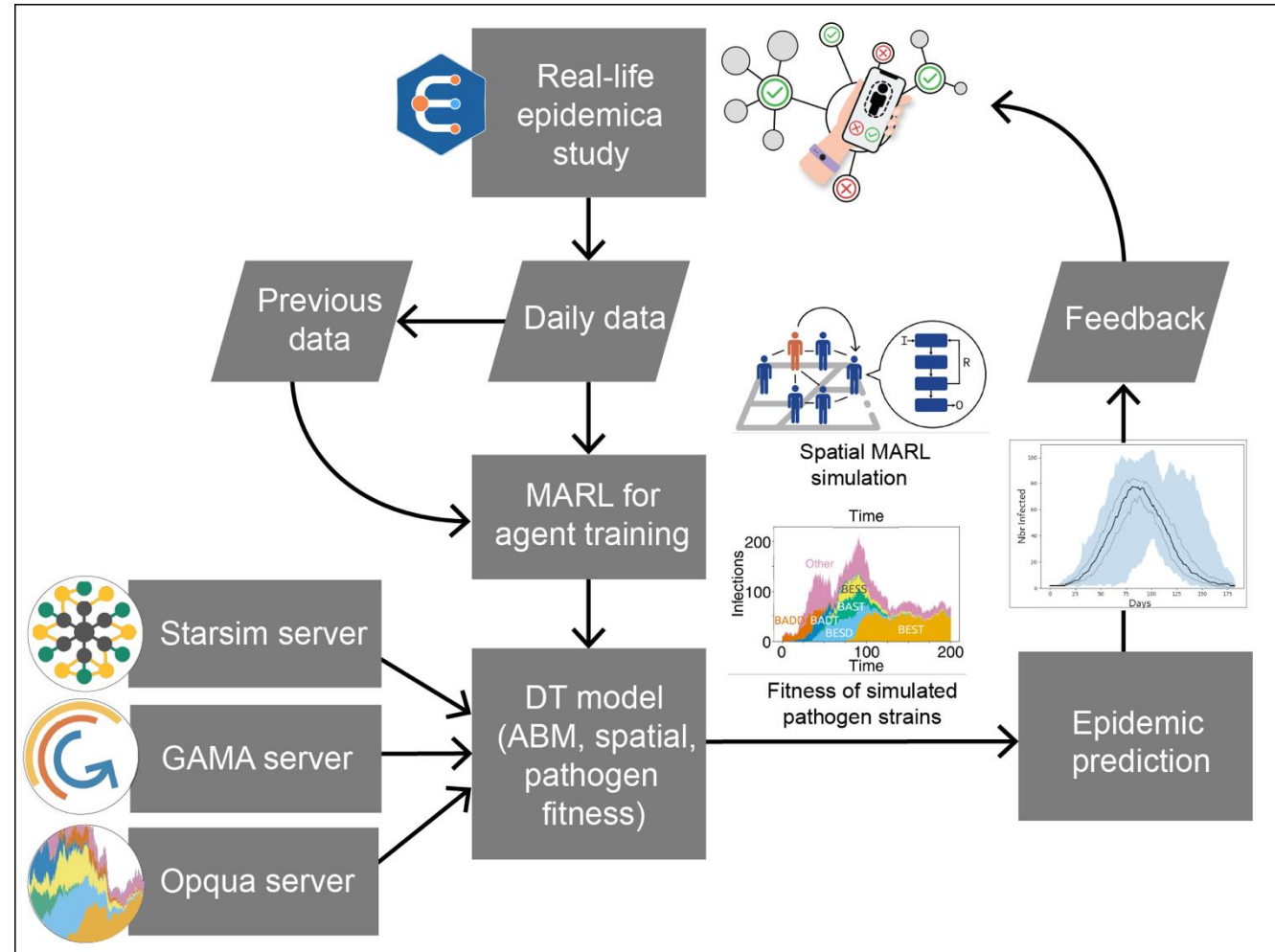


Diagram from an earlier proposal, now we can implement this!

A working proof-of-concept of Epigames v2

Mobile app, cloud server, Starsim twin model, interactive network visualization



* User interface is a placeholder for now

Agentic-ready by design

The same contracts-first discipline extended to AI coding agents

The problem with agent-assisted development

- An agent has no memory of the debugging session that established a rule
- Convention re-learned by trial and error = the exact failure mode contracts exist to prevent
- Switching agents mid-feature → inconsistent assumptions → bugs at the seams
- Long chat history accumulates stale context that silently corrupts new code

Epidemica's approach

- Extensively documented context in the project — what cannot be derived from the code
- AGENTS.md / CLAUDE.md: machine-readable convention files agents read before acting
- Agent skills: named, invokable procedures for recurring workflows (test sweeps, regression checks)
- License scanner + transcript renderer: tooling for responsible agent-assisted authorship

Contracts before code — whether the reader is a contributor, a collaborator, or a language model

Next steps

We can build on top of a strong foundation

Available now

- First field epigame pilot — 7-day “quarantine” game
- Contact logging observational studies
- Data export + analysis validation

Next 6 months

- Survey / instrument module (EMA, DCE)
- GraphQL query API for researcher dashboard
- Rotating BLE identifiers (DPIA-ready)
- Multi-channel reach (SMS, email, web)

Later (1 – 2 years)

- Biosensing module (lateral flow, wearables)
- Environmental sensing (air quality, ventilation)
- CIAS integration for complex interventions
- FAIR metadata and cross-site pooling

Get involved!

1 Run a study

Define a protocol bundle,
deploy on your infrastructure,
collect standardized data

2 Build a Module

Add a new measurement
capability — biosensing,
environmental, behavioral

3 Join a Pilot

Participate in an epigame
or plan a site-specific
feasibility study

contact@epidemica.info · <https://epidemica.info>

Epidemica

Open source · Self-hostable · GDPR-compliant

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